

DSBDA Oral Examination

Data Science & Big Data Analytics

Complete Question & Answer Guide

Experiments 1 – 10 | General Concepts | Diagram-Based Questions

Coverage	Experiments 1–10 + General Concepts
Focus	Basic concepts, code logic, diagram questions
Purpose	Oral / Viva preparation — Base level passer

SECTION 1: General Concepts — Must Know for Any Experiment

Q: What is Data Science?

A: Data Science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data. It combines statistics, programming, and domain expertise.

Q: What is the difference between Data Science and Big Data Analytics?

A: Data Science focuses on extracting insights using statistics and ML from any size of data. Big Data Analytics specifically deals with very large datasets (Volume, Velocity, Variety — 3 Vs) that cannot be processed by traditional tools.

Q: What is a DataFrame in pandas?

A: A DataFrame is a 2-dimensional labeled data structure in pandas, similar to a spreadsheet or SQL table. It has rows and columns, where each column can hold different data types.

Q: What is the difference between iloc and loc?

A: loc is label-based indexing (uses row/column names), while iloc is integer-position-based indexing (uses numeric index 0, 1, 2...). Example: `df.loc['a']` vs `df.iloc[0]`.

Q: What is missing data? How do you handle it?

A: Missing data are NaN/null values in a dataset. Handling techniques: (1) Drop rows/columns using `dropna()`, (2) Fill with mean/median/mode using `fillna()`, (3) Forward/backward fill, (4) Predict missing values using ML models.

Q: What is the difference between supervised and unsupervised learning?

A: Supervised learning uses labeled training data (e.g., Linear Regression, Logistic Regression, Naive Bayes). Unsupervised learning finds hidden patterns without labels (e.g., K-Means Clustering, PCA).

Q: What is overfitting and underfitting?

A: Overfitting: model learns training data too well including noise — poor test performance. Underfitting: model is too simple — poor training and test performance. Solution: cross-validation, regularization, more data.

Q: What is train-test split? Why is it important?

A: We split data into training set (model learns from it) and test set (model is evaluated on it). This checks how well the model generalizes to unseen data. Typical split: 70-80% train, 20-30% test.

Q: What is the role of sklearn (scikit-learn)?

A: Scikit-learn is a Python ML library providing tools for classification, regression, clustering, preprocessing, model selection, and evaluation — all with a consistent fit/predict API.

Q: What is normalization vs standardization?

A: Normalization scales data to [0,1] range using Min-Max scaling. Standardization transforms data to have mean=0 and std=1 (Z-score). Standardization is preferred when data has outliers.

Q: What is a confusion matrix?

A: A confusion matrix is a table showing TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative) — used to evaluate classification model performance.

Q: Define Accuracy, Precision, Recall, and F1-Score.

A: Accuracy = $(TP+TN)/(TP+TN+FP+FN)$. Precision = $TP/(TP+FP)$ — how many predicted positives are correct. Recall = $TP/(TP+FN)$ — how many actual positives are found. F1 = $2*(Precision*Recall)/(Precision+Recall)$ — harmonic mean of both.

Q: What is the purpose of the describe() function?

A: `df.describe()` gives summary statistics for numeric columns: count, mean, std, min, 25th/50th/75th percentile, and max. Useful for quick data overview.

Q: What is Label Encoding vs One-Hot Encoding?

A: Label Encoding assigns integer values to categories (0,1,2...) — good for ordinal data. One-Hot Encoding creates binary columns for each category (`get_dummies`) — good for nominal data to avoid ordinal assumptions.

Q: What is the difference between a bar chart, histogram, and boxplot?

A: Bar chart: compares categories (discrete X). Histogram: shows frequency distribution of continuous data (bins). Boxplot: shows median, quartiles, and outliers — good for spread and skewness.

SECTION 2: Experiment-Wise Questions (1–10)

Experiment 1 — Data Wrangling I (Data Preprocessing & Encoding)

Q: What is Data Wrangling?

A: Data Wrangling (also called Data Munging) is the process of cleaning, transforming, and preparing raw data into a usable format for analysis. It includes handling missing values, type conversions, and encoding.

Q: What does `isnull().sum()` do?

A: `isnull()` returns a boolean DataFrame where True means the value is missing (NaN). `.sum()` counts the number of True values per column, giving the total missing values in each column.

Q: What is the purpose of `dropna()`?

A: `dropna()` removes rows (or columns with `axis=1`) that contain NaN values. It's used to clean the dataset by eliminating incomplete records.

Q: What is `LabelEncoder`? When is it used?

A: `LabelEncoder` from `sklearn` converts categorical text labels into numeric values (0, 1, 2...). It is used when an ML model requires numeric input. Best for ordinal (ordered) categories.

Q: What is `pd.get_dummies()`? How is it different from `LabelEncoder`?

A: `get_dummies()` performs One-Hot Encoding — creates a new binary (0/1) column for each unique category value. Unlike `LabelEncoder`, it doesn't impose any ordinal relationship between categories.

Q: What does `value_counts()` return?

A: `value_counts()` returns the count of each unique value in a Series, sorted in descending order. Useful to understand the distribution of categorical variables.

Q: What is `pd.concat()`?

A: `pd.concat()` joins DataFrames along rows (`axis=0`) or columns (`axis=1`). In the experiment, it adds the encoded dummy columns alongside the original DataFrame.

Q: DIAGRAM Q: Draw the Data Preprocessing Pipeline.

A: Raw Data → Load (`pd.read_csv`) → Check Shape (`df.shape`) → Check Types (`df.dtypes`) → Handle Missing (`isnull/dropna/fillna`) → Encode Categorical (`LabelEncoder/get_dummies`) → Normalize/Scale → Clean Dataset

■ Remember this as: Load → Inspect → Clean → Encode → Scale

Experiment 2 — Data Wrangling II (Outliers & Transformation)

Q: What is an outlier?

A: An outlier is a data point that significantly deviates from the rest of the dataset. It can distort statistical analysis and model training.

Q: What is IQR? How is it used to detect outliers?

A: IQR (Interquartile Range) = $Q3 - Q1$ (75th percentile - 25th percentile). Outliers are values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. This is the standard Tukey method.

Q: What are `np.percentile()` parameters?

A: `np.percentile(dataset, [25, 75])` returns the 25th percentile ($Q1$) and 75th percentile ($Q3$) of the dataset — used to compute IQR for outlier detection.

Q: What is data transformation? Give examples.

A: Data transformation changes the scale or distribution of a variable. Examples: Log transformation (reduces right skew), Square root, Min-Max normalization, Standardization (Z-score), and date parsing like extracting year from datetime.

Q: What is `pd.DatetimeIndex`? Why is it used?

A: `pd.DatetimeIndex` converts a column to datetime format, allowing extraction of components like year, month, day. In the experiment, it extracts the admission year to compute years elapsed.

Q: What is a lambda function? Give an example.

A: A lambda is an anonymous single-line function. Example: `df['Fees_M$'] = df['Fees$'].apply(lambda x: round(x/1000000))` — divides each fee value by 1 million and rounds it.

Q: DIAGRAM Q: Draw the IQR Outlier Detection diagram.

A: Number line: [Lower Bound] --- [Q1] --- [Median] --- [Q3] --- [Upper Bound]. Values outside Lower/Upper bounds are outliers. Lower = $Q1 - 1.5 \times IQR$, Upper = $Q3 + 1.5 \times IQR$.

Experiment 3 — Descriptive Statistics (Central Tendency & Variability)

Q: What are Measures of Central Tendency?

A: Mean: arithmetic average. Median: middle value when sorted — robust to outliers. Mode: most frequently occurring value. They describe the center of a distribution.

Q: What is Standard Deviation?

A: Standard Deviation measures the spread/dispersion of data from the mean. A low SD means data is clustered near the mean; a high SD means data is spread out.

Q: What is the difference between mean and median? When is median preferred?

A: Mean is affected by outliers; median is not. Median is preferred for skewed distributions (e.g., income data) where extreme values would distort the mean.

Q: What does `groupby()` do in pandas?

A: `groupby()` splits the DataFrame into groups based on a column's values, then you can apply aggregate functions (mean, sum, count) to each group. Useful for group-level statistics.

Q: What is the `scipy.stats.mode()` function?

A: `stats.mode()` returns the mode (most common value) and its count from an array. It is from the SciPy library and returns a `ModeResult` object with `.mode` and `.count` attributes.

Q: What is the Iris dataset?

A: Iris is a classic ML dataset with 150 records of iris flowers across 3 species (Setosa, Versicolor, Virginica). Features: SepalLength, SepalWidth, PetalLength, PetalWidth — all numeric.

Q: DIAGRAM Q: Draw a Normal Distribution and mark Mean, Median, Mode.

A: Bell-shaped curve: In a perfect normal distribution, Mean = Median = Mode, all at the peak center. 68% data within 1 SD, 95% within 2 SD, 99.7% within 3 SD (Empirical Rule).

Experiment 4 — Data Analytics I: Linear Regression (Boston Housing)

Q: What is Linear Regression?

A: Linear Regression models the relationship between a dependent variable (Y) and one or more independent variables (X) by fitting a straight line: $Y = mX + c$. It minimizes the sum of squared errors.

Q: What is the Boston Housing Dataset?

A: It contains information on 506 Boston suburbs with 13 features (CRIM, ZN, INDUS, CHAS, NOX, RM, AGE, DIS, RAD, TAX, PTRATIO, B, LSTAT) used to predict MEDV (median house value).

Q: What is the difference between fit() and predict()?

A: `fit(X_train, y_train)` trains the model by learning coefficients from training data. `predict(X_test)` uses the learned model to make predictions on new/unseen data.

Q: What do reg.coef_ represent?

A: `reg.coef_` contains the coefficients (slopes) for each feature in the linear regression equation. Each value represents the change in the target variable for a one-unit change in that feature.

Q: What is Mean Squared Error (MSE)?

A: MSE = average of squared differences between predicted and actual values. Lower MSE = better model. It penalizes large errors more than small ones.

Q: What is the difference between simple and multiple linear regression?

A: Simple Linear Regression: one independent variable ($Y = mX + c$). Multiple Linear Regression: two or more independent variables ($Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$).

Q: What is R-squared (R2 score)?

A: R2 score indicates how much variance in the dependent variable is explained by the model. $R^2 = 1$ means perfect fit; $R^2 = 0$ means model explains nothing. Ranges from 0 to 1.

Q: DIAGRAM Q: Draw a Simple Linear Regression plot.

A: X-axis: independent variable (e.g., number of rooms). Y-axis: dependent variable (house price). Plot scattered data points and a best-fit straight line through them. The line minimizes vertical distances from points (residuals).

Experiment 5 — Data Analytics II: Logistic Regression + Confusion Matrix

Q: What is Logistic Regression? How is it different from Linear Regression?

A: Logistic Regression is used for classification (output is a class 0/1), not regression. It uses the sigmoid function to output probabilities between 0 and 1. Linear Regression predicts continuous values.

Q: What is the Sigmoid function?

A: Sigmoid: $f(x) = 1 / (1 + e^{-x})$. It maps any value to (0,1), giving probability. If output > 0.5, class = 1; else class = 0.

Q: What is StandardScaler? Why is it used before Logistic Regression?

A: StandardScaler standardizes features by removing mean and scaling to unit variance. Logistic Regression (and many ML models) perform better when features are on the same scale.

Q: Explain the Confusion Matrix with TP, TN, FP, FN.

A: TP: predicted positive, actually positive. TN: predicted negative, actually negative. FP: predicted positive, actually negative (Type I error). FN: predicted negative, actually positive (Type II error).

Q: How do you calculate Accuracy and Error Rate?

A: Accuracy = $(TP + TN) / \text{Total}$. Error Rate = $(FP + FN) / \text{Total} = 1 - \text{Accuracy}$. In the experiment: Accuracy = $(74+31)/120 = 0.875$ (87.5%).

Q: What is Precision vs Recall? When to use which?

A: Precision = $TP/(TP+FP)$ — use when FP cost is high (e.g., spam detection). Recall = $TP/(TP+FN)$ — use when FN cost is high (e.g., cancer detection). F1-score balances both.

Q: DIAGRAM Q: Draw the Confusion Matrix grid.

A: 2x2 table: Rows = Actual (Positive, Negative), Columns = Predicted (Positive, Negative). Cells: [TP, FP] / [FN, TN]. Diagonal (TP, TN) = correct predictions. Off-diagonal = errors.

Experiment 6 — Data Analytics III: Naive Bayes Classification

Q: What is the Naive Bayes algorithm?

A: Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes all features are independent (the 'naive' assumption). $P(\text{Class}|\text{Features}) \propto P(\text{Features}|\text{Class}) * P(\text{Class})$.

Q: What is Bayes' Theorem?

A: $P(A|B) = P(B|A) * P(A) / P(B)$. In classification: $P(\text{class}|\text{data}) = P(\text{data}|\text{class}) * P(\text{class}) / P(\text{data})$. We pick the class with the highest posterior probability.

Q: What is Gaussian Naive Bayes? Why is it used for Iris?

A: GaussianNB assumes continuous features follow a normal (Gaussian) distribution. Since Iris features (sepal/petal measurements) are continuous, GaussianNB is appropriate.

Q: What is a Violin Plot? How is it different from a Boxplot?

A: A violin plot combines a box plot with a kernel density estimation (KDE) showing the full distribution shape. Boxplot only shows quartiles and outliers; violin plot shows where data is concentrated.

Q: What is a Heatmap? What does `sns.heatmap(iris.corr())` show?

A: A heatmap uses color to represent values in a matrix. `iris.corr()` computes the correlation matrix between all numeric features. The heatmap shows which features are strongly correlated (+1) or anti-correlated (-1).

Q: What is correlation? What do values +1, 0, -1 mean?

A: +1: perfect positive correlation (both increase together). 0: no correlation. -1: perfect negative correlation (one increases, other decreases). PetalLength and PetalWidth in Iris have high positive correlation (~0.96).

Q: DIAGRAM Q: Draw and explain a Heatmap concept.

A: Grid where rows and columns are features. Each cell shows correlation value. Color scale: dark red/blue = strong correlation, white/light = weak. Diagonal is always 1.0 (feature correlated with itself).

Experiment 7 — Text Analytics (NLP: Tokenization, POS, TF-IDF)

Q: What is NLP (Natural Language Processing)?

A: NLP is a branch of AI that enables computers to understand, interpret, and generate human language. It includes tasks like tokenization, POS tagging, sentiment analysis, and text classification.

Q: What is Tokenization? What are its types?

A: Tokenization splits text into smaller units called tokens. Sentence tokenization (`sent_tokenize`) splits into sentences. Word tokenization (`word_tokenize`) splits into individual words.

Q: What is POS Tagging?

A: Part-of-Speech (POS) tagging labels each word with its grammatical role: NN (noun), VB (verb), JJ (adjective), RB (adverb), DT (determiner), etc. Used for grammar analysis and NER.

Q: What are Stop Words? Why are they removed?

A: Stop words are common, low-information words like 'is', 'the', 'a', 'and'. They are removed to reduce noise and focus on meaningful words for analysis or modeling.

Q: What is Stemming vs Lemmatization?

A: Stemming: crudely chops word endings (running → run, better → bet). Lemmatization: reduces to root dictionary form using vocabulary (running → run, better → good). Lemmatization is more accurate.

Q: What is TF-IDF? Explain TF and IDF separately.

A: TF (Term Frequency) = count of word in document / total words in document. IDF (Inverse Document Frequency) = $\log(N / \text{number of documents containing word})$. TF-IDF = TF * IDF. High value = word is important and unique to that document.

Q: What is PorterStemmer in NLTK?

A: PorterStemmer is an algorithm in NLTK that applies a series of suffix-stripping rules to reduce words to their stem. Example: 'learning' → 'learn', 'originated' → 'origin'.

Q: DIAGRAM Q: Draw the NLP Text Preprocessing Pipeline.

A: Raw Text → Sentence Tokenization → Word Tokenization → Lowercasing → Stop Word Removal → Stemming/Lemmatization → POS Tagging → Clean Tokens → TF-IDF Vectorization → Feature Matrix

Experiment 8 — Data Visualization I: Titanic Dataset (Seaborn)

Q: What is Seaborn? How is it different from Matplotlib?

A: Seaborn is a high-level visualization library built on Matplotlib. It provides attractive statistical plots with less code and works naturally with pandas DataFrames. Matplotlib gives more low-level control.

Q: What is `sns.distplot()` / `sns.histplot()`?

A: `distplot` (deprecated) / `histplot` shows the distribution of a numeric variable. It plots a histogram and optionally a KDE (Kernel Density Estimate) curve showing the probability density.

Q: What is `sns.jointplot()`?

A: `jointplot` shows the relationship between two variables with scatter plot in the center and individual distributions (histograms/KDE) on the margins. Good for bivariate analysis.

Q: What is the Titanic dataset?

A: Titanic dataset has 891 rows with passenger info: survived (0/1), pclass, sex, age, fare, embarked, etc. Used to analyze survival patterns — women and higher-class passengers had better survival rates.

Q: What does `kde=False` in `distplot` do?

A: `kde=False` removes the Kernel Density Estimate curve, showing only the histogram bars without the smooth density line overlay.

Q: What is a rugplot?

A: A rugplot draws a small vertical tick mark at each data point along the x-axis. It shows the exact locations of all data values, useful for identifying clusters and gaps.

Q: DIAGRAM Q: What does a skewed histogram look like?

A: Right-skewed (positive skew): long tail on the right, peak on left — e.g., fare distribution in Titanic (most fares are low, few are very high). Left-skewed: tail on left, peak on right.

Experiment 9 — Data Visualization II: Boxplot (Titanic: Age vs Gender)

Q: What is a Boxplot and what does it display?

A: A boxplot shows: Minimum, Q1 (25th percentile), Median (Q2/50th), Q3 (75th percentile), Maximum, and Outliers (as individual dots). The box spans $IQR = Q3 - Q1$.

Q: What does `sns.boxplot(x='sex', y='age', hue='survived')` show?

A: It shows the age distribution for males and females, split further by survival status (0 or 1). This lets you compare age distributions across 4 groups simultaneously.

Q: What observations can you make from the Titanic age-gender boxplot?

A: Females who survived tend to be slightly younger. Males who didn't survive have a wider age spread. Children (low age) had higher survival chances. The median age of survivors vs non-survivors differs.

Q: What is `sns.violinplot`? How does it compare to boxplot?

A: A violin plot shows the full distribution shape (KDE) + boxplot summary. It reveals bimodal or multimodal distributions that a boxplot would hide. Wider sections = more data density there.

Q: What is `sns.stripplot`?

A: A stripplot draws each individual data point as a dot along a categorical axis. It shows all individual observations and their distribution within each category.

Q: DIAGRAM Q: Label all parts of a Boxplot.

A: Draw a rectangle (box) with a line at median. Whiskers extend to min/max (or $1.5 \times IQR$). Dots beyond whiskers = outliers. Left/bottom of box = Q1. Right/top of box = Q3. Box width = IQR.

Experiment 10 — Data Visualization III: Iris Dataset (Histograms & Boxplots)

Q: What are the features of the Iris dataset and their types?

A: `SepalLengthCm`, `SepalWidthCm`, `PetalLengthCm`, `PetalWidthCm` — all numeric (continuous). `Species` — nominal (categorical): `Iris-setosa`, `Iris-versicolor`, `Iris-virginica`.

Q: How do histograms help in understanding feature distributions?

A: Histograms show the frequency (count) of values in bins. They reveal if data is normally distributed, skewed, bimodal, or has gaps. Each Iris feature has a distinct distribution shape.

Q: What outlier pattern can you see in Iris sepal width boxplot?

A: `SepalWidthCm` has several outliers above the upper whisker, indicating a few samples with unusually wide sepals — these are likely `Iris setosa` which has wider sepals than other species.

Q: What is `sns.histplot()` syntax?

A: `sns.histplot(data=df, x='column_name')` — plots a histogram of the specified column from the DataFrame. Optional parameters: `bins`, `kde=True` (adds density curve), `hue` (color by category).

Q: What does `data.describe()` tell you about the Iris dataset?

A: For each numeric feature: count=150, mean, std, min, 25%, 50%, 75%, max. PetalLength has higher std than SepalWidth — more variability. Helps spot features with large ranges that may need scaling.

Q: How can you compare feature distributions across species?

A: Use `sns.boxplot(x='Species', y='Feature')` or violin plots. Setosa is easily separable (smaller petals). Versicolor and Virginica overlap more, making classification harder.

Q: DIAGRAM Q: Draw and compare histograms of SepalLength vs PetalLength (conceptual).

A: SepalLength histogram: roughly normal/bell shaped, range ~4.3–7.9cm. PetalLength histogram: bimodal — one peak ~1.5cm (Setosa) and another ~4-5cm (Versicolor/Virginica). This bimodal shape is a key insight showing species separation.

SECTION 3: Quick Revision — Key Functions Cheatsheet

Function / Concept	Purpose	Experiment
pd.read_csv()	Load CSV into DataFrame	1,2,3...
df.isnull().sum()	Count missing values per column	1,2
df.describe()	Summary statistics of numeric cols	1,3,9,10
df.dtypes	Check data types of each column	1
df.dropna()	Remove rows/cols with NaN values	1
LabelEncoder	Encode categories to integers	1,6
pd.get_dummies()	One-Hot Encode categorical column	1
np.percentile([],25/75)	Calculate Q1, Q3 for IQR	2
IQR = Q3 - Q1	Interquartile range for outliers	2
train_test_split()	Split data into train and test sets	4,5,6
LinearRegression()	Fit linear regression model	4
reg.fit(X,y)	Train the model	4,5,6
reg.predict(X)	Make predictions	4,5,6
LogisticRegression()	Classification using sigmoid	5
StandardScaler()	Standardize features (mean=0, std=1)	5
confusion_matrix()	TP/TN/FP/FN matrix	5,6
accuracy_score()	Overall correct predictions / total	5,6
precision_score()	TP / (TP + FP)	5,6
recall_score()	TP / (TP + FN)	5,6
f1_score()	Harmonic mean of Precision & Recall	5,6
GaussianNB()	Naive Bayes for continuous features	6
sns.heatmap(df.corr())	Correlation heatmap	6
nltk.word_tokenize()	Split text into word tokens	7
nltk.pos_tag()	Assign POS tags to tokens	7
PorterStemmer()	Stem words to root form	7
WordNetLemmatizer()	Lemmatize words using WordNet	7
TfidfVectorizer()	Compute TF-IDF feature matrix	7
sns.distplot/histplot()	Histogram + KDE of distribution	8
sns.boxplot()	Boxplot showing quartiles & outliers	9,10
sns.violinplot()	Distribution shape + box summary	6,9

SECTION 4: Common Diagram Questions — Know These!

■ Data Science Process Flow

Business Problem → Data Collection → Data Wrangling → EDA → Model Building → Model Evaluation → Deployment → Monitoring

■ Machine Learning Workflow

Raw Data → Preprocessing → Feature Engineering → Train/Test Split → Model Training → Prediction → Evaluation (Accuracy/F1) → Tune Hyperparameters

■ Confusion Matrix (Binary)

| Predicted + | Predicted - | Actual + | TP | FN | Actual - | FP | TN
Accuracy=(TP+TN)/Total, Precision=TP/(TP+FP), Recall=TP/(TP+FN)

■ Linear Regression Diagram

Scatter plot of X vs Y with best-fit line $Y=mX+c$. Residuals = vertical distance from each point to line. Goal: minimize sum of squared residuals.

■ Logistic Regression Sigmoid

S-shaped curve: X-axis = linear combination (z), Y-axis = probability (0 to 1). Threshold at 0.5: above = Class 1, below = Class 0.

■ TF-IDF Concept

TF = word_count/doc_length (how often word appears). IDF = $\log(N/df)$ (how rare across docs). TF-IDF = TF * IDF. Common words get low IDF; rare important words get high TF-IDF.

■ Boxplot Anatomy

Min --- [Q1---Median---Q3] --- Max, with dots beyond whiskers as outliers. Whisker length = $1.5 * IQR$. Box = $IQR = Q3 - Q1$.

■ IQR Outlier Method

$Q1=25\text{th percentile}$, $Q3=75\text{th percentile}$, $IQR=Q3-Q1$. Lower fence = $Q1-1.5*IQR$. Upper fence = $Q3+1.5*IQR$. Values outside fences are outliers.
